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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Using the image already uploaded to Colab
img = cv2.imread('image1.jpg', 0)

if img is not None:
    # Apply Sobel edge detection
    sobel_x = cv2.Sobel(img, cv2.CV_8U, 1, 0, ksize=5)

    # Save and display the result
    cv2.imwrite('sobel_x.jpg', sobel_x)
    print("Sobel X Result:")
    cv2_imshow(sobel_x)
else:
    print("Error: 'image1.jpg' not found. Please upload it again if it was
```

Sobel X Result:

